Sai Deepak Sana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.79
Narayana Junior College	Class XII (TSBIE)	2023	97.50%
Narayana Olympiad School	Class X (SSC)	2021	100.00%

EXPERIENCE

Member of Technical Staff Intern - Nutanix

- Built **Python-based UI automation** for Nutanix Prism Central Projects 2.0 using Playwright and NuTest, covering 8 end-to-end workflows spanning networking, IAM/RBAC, compute, storage, and category management.
- Engineered a **concurrent browser execution framework** using **asyncio**, browser pooling, and isolated sessions to execute multiple user workflows in parallel while handling asynchronous UI state, rate limits, and per-session failures.
- Designed reusable **Page Object**, **operation**, and **workflow abstractions** with idempotent API-based setup and observed-state verification; generalized subnet creation through shared base components for **VLAN** and **overlay** networking.
- Integrated the workflows into a **multi-user end-to-end** system, implementing bounded waits, exponential backoff, and structured failure diagnostics to ensure reliable resource lifecycles, RBAC visibility, and concurrent execution.

PROJECTS

Chunkstore -- Distributed Content-Addressed Storage (Go, SQLite, TCP)

- Built a distributed content-addressed storage system in Go using FastCDC chunking, SHA-256 deduplication, SQLite metadata, and streaming TCP; transferred only 11.2% of file bytes vs. 98.0% with fixed-size chunking.
- Implemented chain replication, resumable uploads, connection pooling, bounded-memory streaming, and retry/failover; achieved 2.9 times higher upload throughput than client fan-out on a bandwidth-constrained link.
- Engineered crash-consistent writes and automatic replica recovery using *fsync*, atomic rename, failure detection, re-replication, corruption quarantine, and fault injection; validated with 125 tests, race detection, and fuzz testing.

FlashStock - Concurrent Reservation Engine (C++, Go, Python)

- Built a C++20 concurrent inventory reservation engine using mutex, cache-line-padded lock striping, and atomic CAS; enforced atomic stock updates to prevent overselling and lost updates under high concurrency.
- Engineered WAL-based durability with CRC32C checks, *fsync*, opportunistic group commit, atomic snapshots, and redo-only crash recovery; validated 10 crash scenarios, corruption handling, and *kill -9* recovery.
- Built a Go HTTP service over Unix sockets with a Python/Go benchmarking harness; achieved 2.9 times throughput with lock striping and 27 times with group commit at 200 clients.

ShopSense -- Multi-Agent Customer Care & Order Operations Assistant

- Developed a multi-agent customer-care assistant for ticket triage, order lookup, cancellation, returns, and refunds using LangGraph orchestration and MCP-based tool integration.
- Built policy-grounded decision making with RAG, customer memory, plan verification, confirmation checks, refund limits, and privilege separation to prevent untrusted customer inputs from triggering state-changing operations.
- Designed an evaluation suite against a single-agent baseline, measuring task success, policy-violation rate, prompt-injection resistance, pass^k reliability, and tool-call efficiency across normal, ambiguous, and adversarial scenarios.

RISC-V Simulator

- Built a C++ RISC-V assembler and simulator that parses assembly into machine code and executes the fetch-decode-execute cycle, updating the register file and simulated memory.
- Implemented a configurable cache model -- capacity, block size, associativity, LRU/FIFO replacement, write-back/write-through policies -- instrumented with hit/miss rates to analyze memory-hierarchy performance tradeoffs.

TECHNICAL SKILLS

- **Languages** - C++, C, Python, Go, SQL
- **Systems & Backend** - Concurrency & Multithreading, Synchronization, Distributed Systems, REST APIs, TCP/IP.
- **Tools & Frameworks** - Linux, Git, Gerrit, GDB, VS Code, Make, LaTeX, asyncio, Playwright.
- **Competitive Programming** - Codeforces, LeetCode, CodeChef, AtCoder

ACHIEVEMENTS

- Selected for Amazon ML Summer School 2026
- Secured State Ranks **119** and **149** in AP & TS EAPCET 2023, out of 238k+ and 195k+ candidates respectively
- Secured an All India Rank **1521** in KVPY (SA) 2021 out of 50k+ candidates

RELEVANT COURSES

- Algorithms, Data Structures, Operating Systems, Computer Networks, Computer Architecture, Machine Learning, Deep Learning, DBMS, Software Engineering, Computational Number Theory, Probability.

POSITION OF RESPONSIBILITIES

- **Design Coordinator** at Finance and Consulting Club, IIT Hyderabad